Material collection for laboratory examination

Blood collection

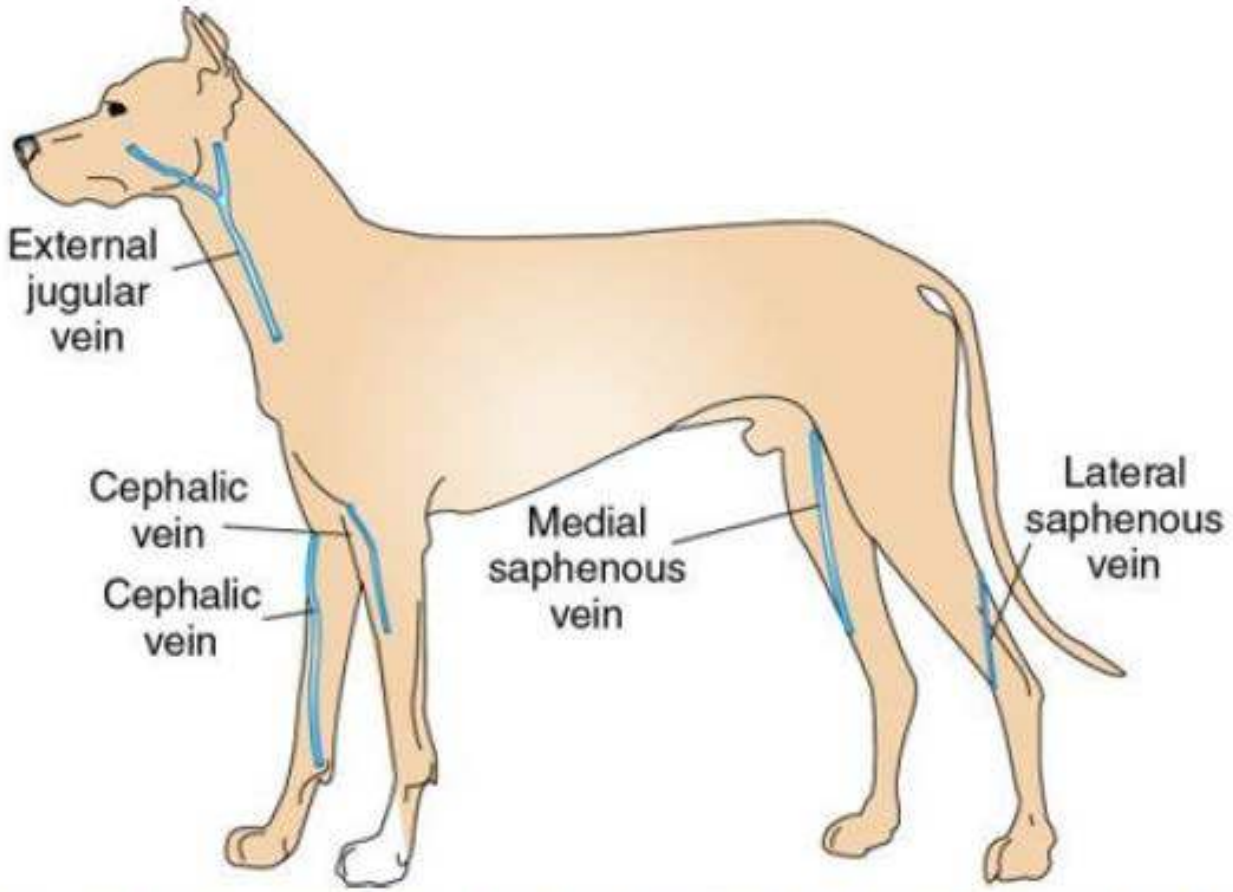
- Venipuncture



Equipment required for venipuncture in dogs and cats.

(Taylor, 2022)

(Taylor, 2022)



The veins accessible for collection of venous blood in dogs and cats.

Jugular vein 1



Restraining a cat for jugular venipuncture.

(Taylor, 2022)

Jugular vein 2

(Taylor, 2022)



Restraining a medium-sized dog for jugular venipuncture.



Restraining a medium-sized dog for jugular venipuncture.

Jugular vein 3

(Taylor, 2022)



Restraining a large dog during jugular venipuncture.



The external jugular vein is located within the jugular furrow, which is a groove on the side of the neck dorsolateral to the trachea.

Jugular vein 4

(Taylor, 2022)





Distending the jugular vein.

Cephalic vein 1



Proper restraint for cephalic venipuncture.

(Taylor, 2022)

Cephalic vein 2

(Taylor, 2022)





Using the thumb of the hand holding the leg, the cephalic vein is rolled laterally and compressed so that it becomes distended with blood.

Cephalic vein 3



Stabilizing the cephalic vein.

(Taylor, 2022)

Lateral saphenous vein 1



Proper restraint for lateral saphenous venipuncture.

(Taylor, 2022)

Lateral saphenous vein 2





Compressing the lateral saphenous vein, causing it to distend with blood.

(Taylor, 2022)

Lateral saphenous vein 3



Stabilizing the lateral saphenous vein by placing the thumb adjacent to the vein during venipuncture.

(Taylor, 2022)

Medial saphenous vein 1

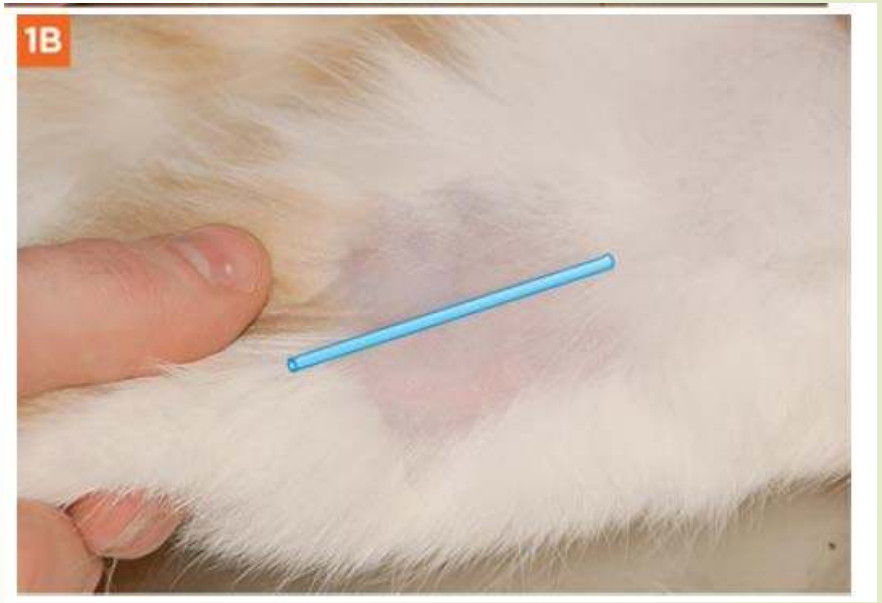


Proper restraint of a cat for medial saphenous venipuncture.

(Taylor, 2022)

Medial saphenous vein 2





Occluding the medial saphenous vein, causing it to distend with blood.

(Taylor, 2022)

Urine collection

- 導尿 Urinary catheterization
  - Male
  - Female
- 膀胱穿刺 Cystocentesis

To collect urine samplesthat are free from contamination by bacteria, cells, and debrisoriginating from the lower urinary tract

Urinary catheterization for male dog 1



(Taylor, 2022)

Urinary catheterization for male dog 2

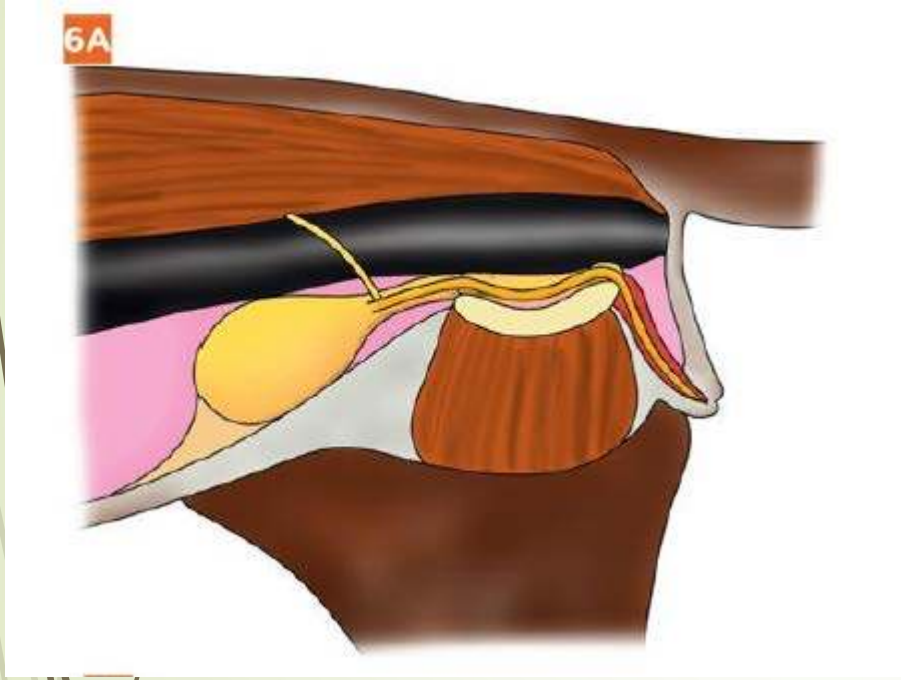


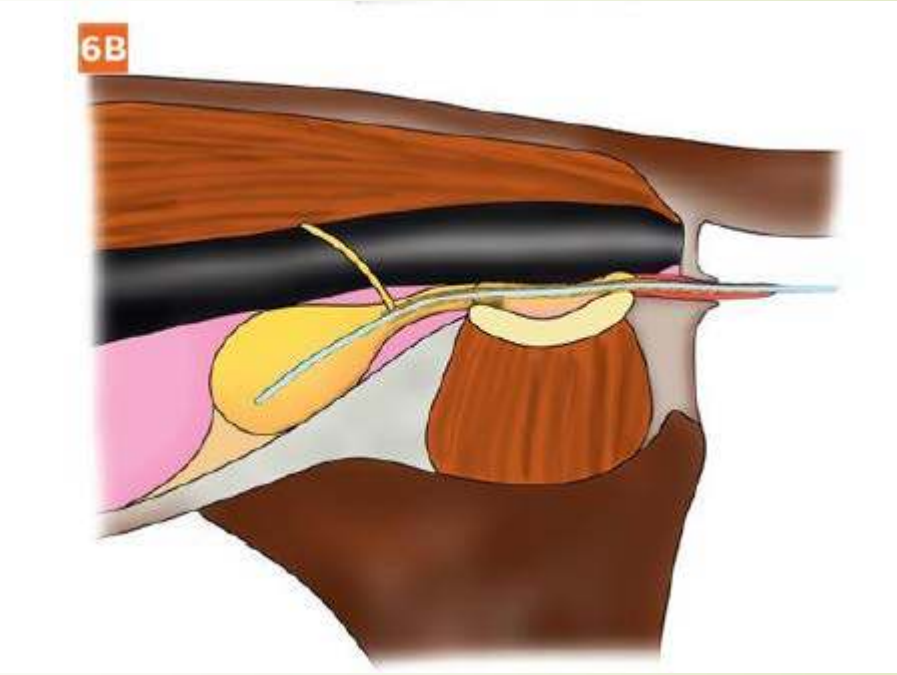


Lubricating the tip of a tomcat urinary catheter.

(Taylor, 2022)

Urinary catheterization for male dog 3

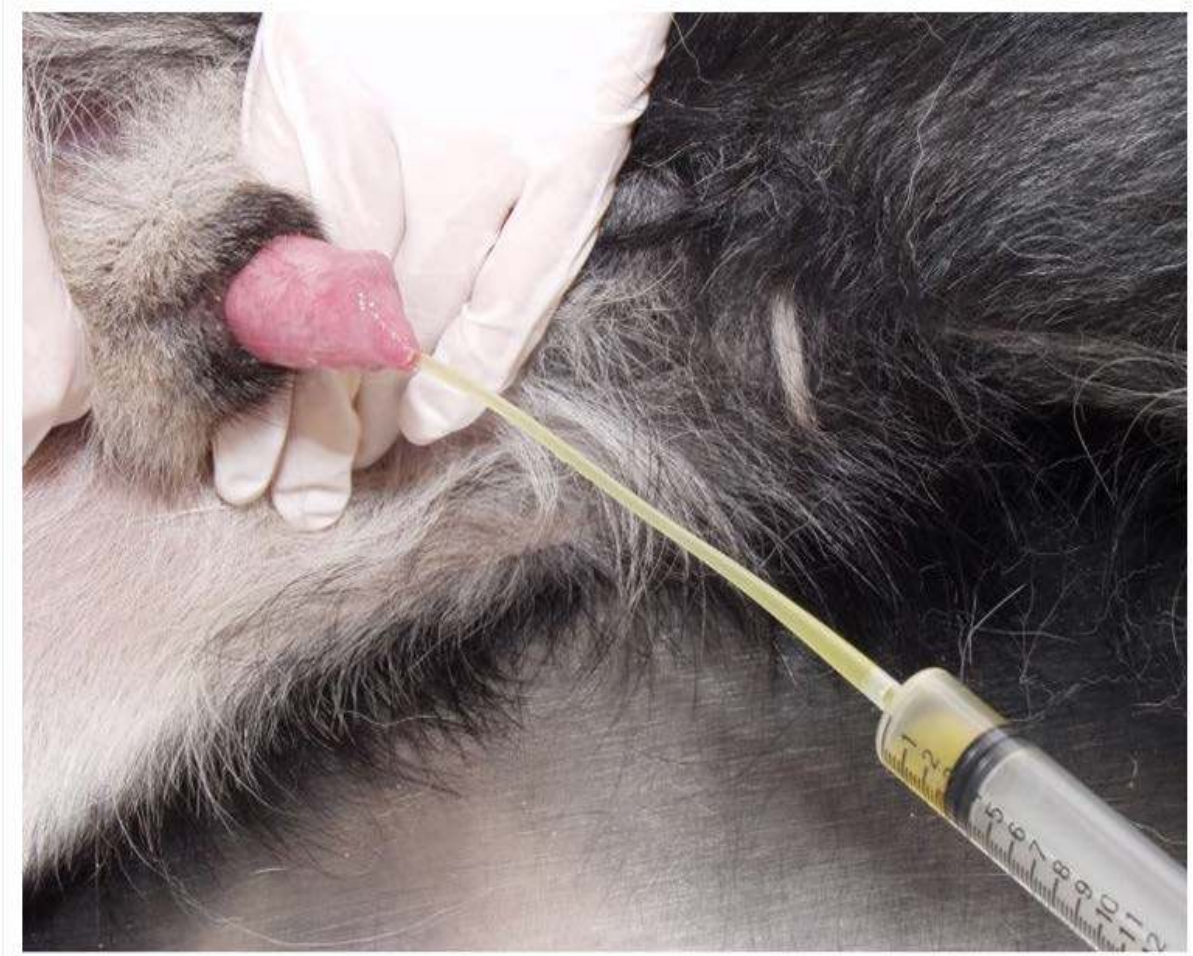




Pulling the extruded penis straight back caudally reduces the natural curvature of the urethra and simplifies passage of a urinary catheter.

(Taylor, 2022)

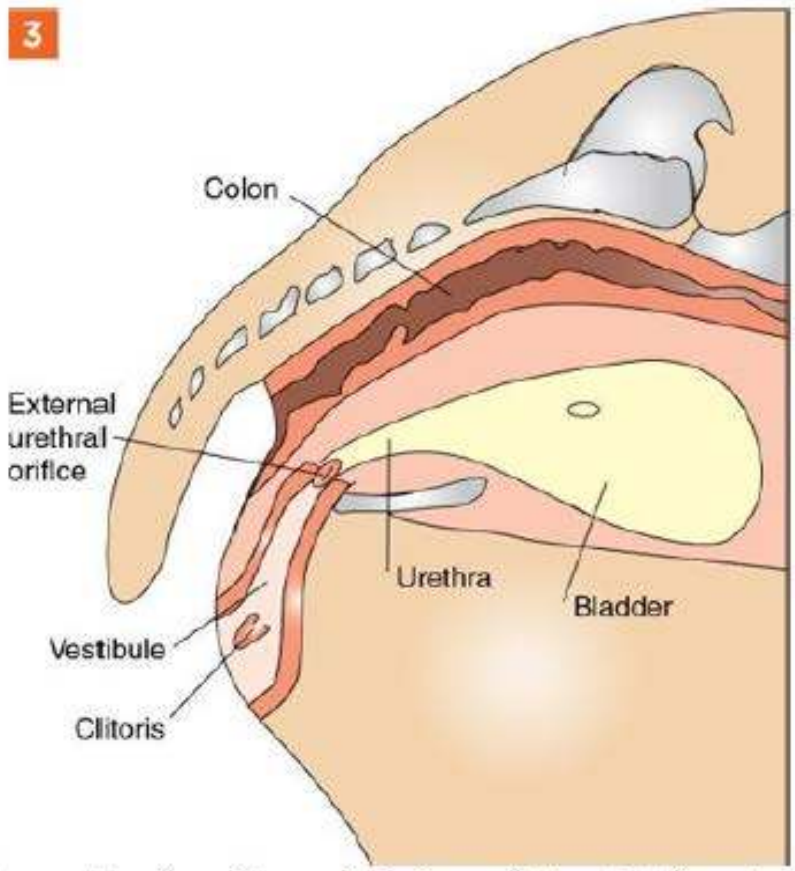
Urinary catheterization for male dog 4



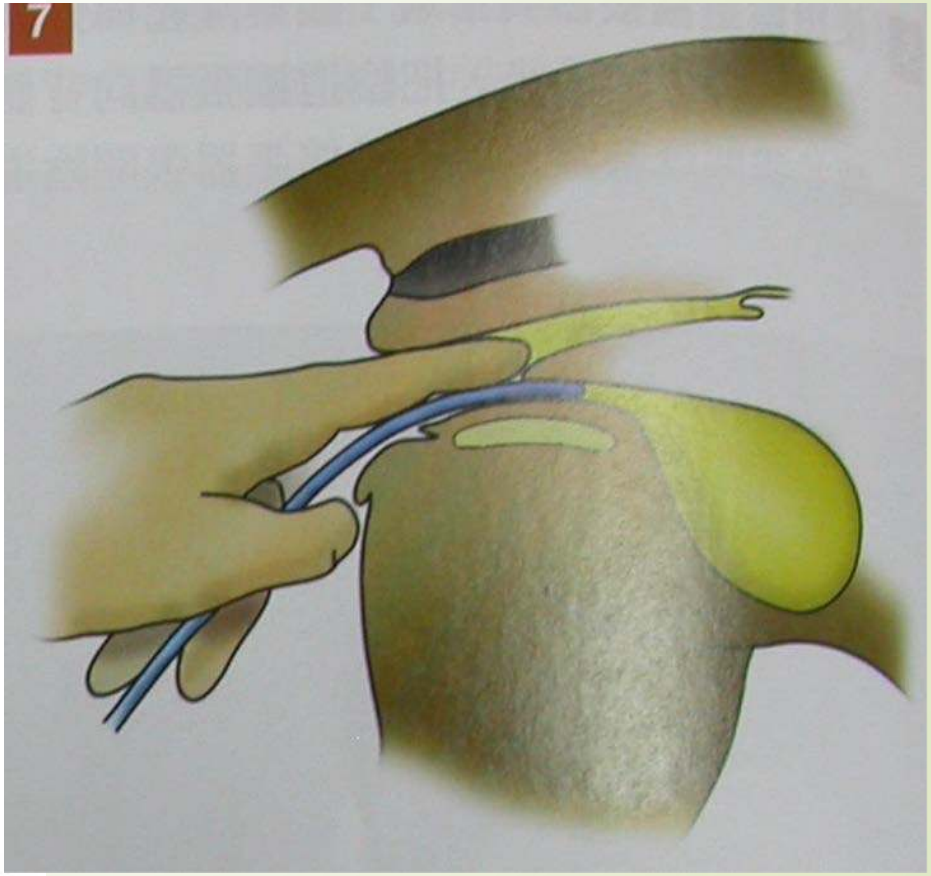
(Taylor, 2022)

Urinary catheterization for female dog 1

(Taylor, 2022)



The caudal portion of the vagina is the vestibule, which is angled steeply dorsocranially until just past the urethral tubercle.



Urinary catheterization for female dog 2



Restraint for urinary catheterization.



The perivulvar skin and vulva are cleaned with antiseptic solution and rinsed.

(Taylor, 2022)

Urinary catheterization for female dog 3

(Taylor, 2022)



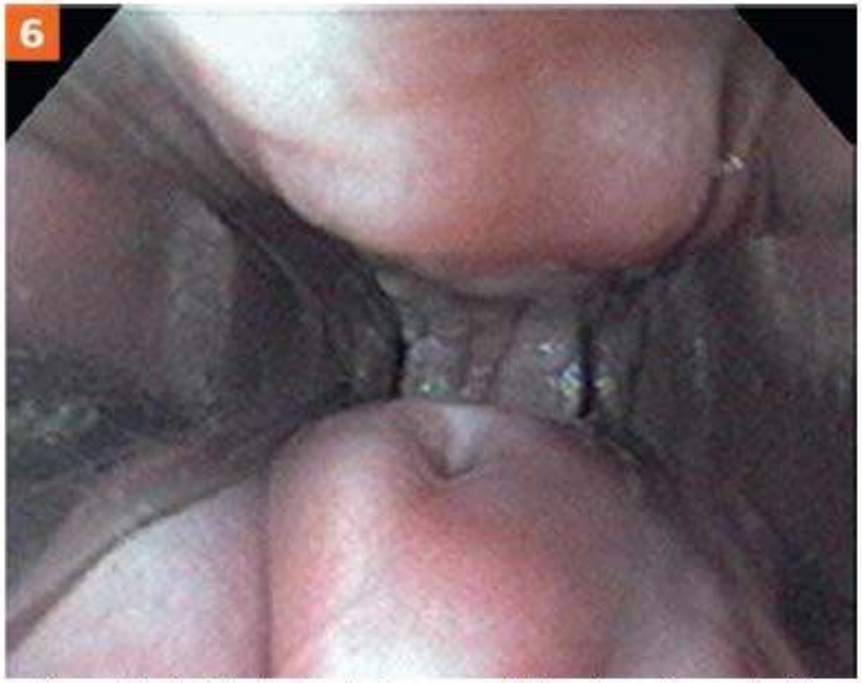
The vagina and vestibule are flushed with sterile saline injected through a syringe.



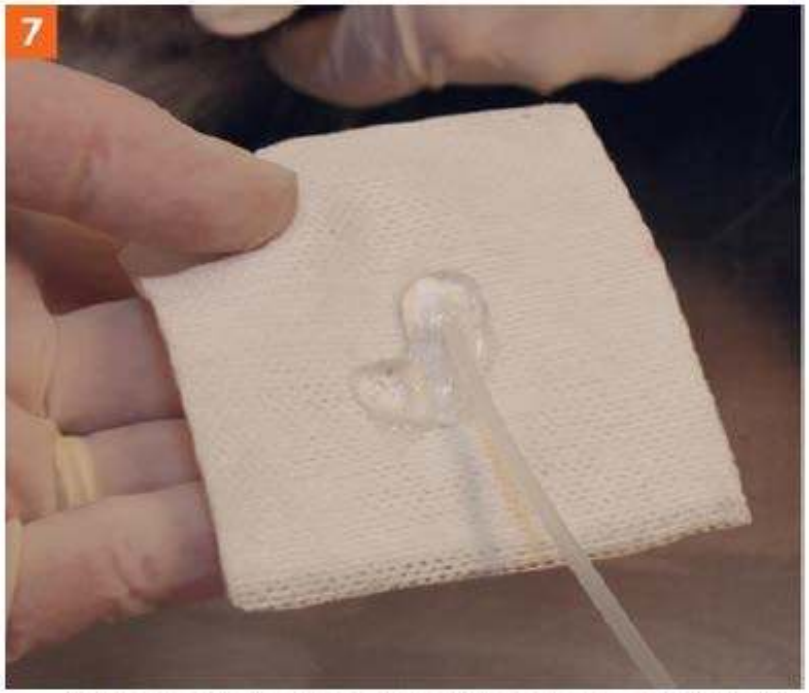


A speculum is directed dorsally after insertion through the vulvar lips, and, once inserted, the wings are spread to visualize the vestibule and vagina.

Urinary catheterization for female dog 4



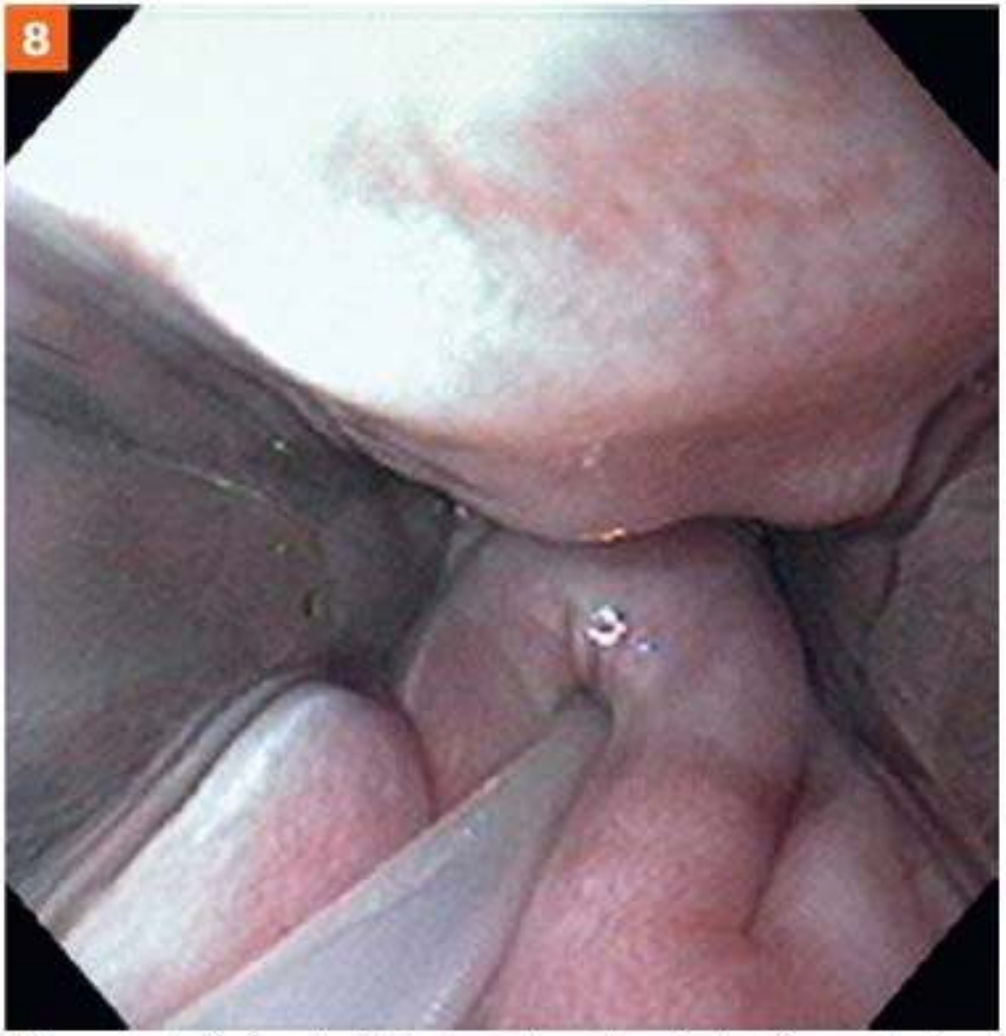
The urethral orifice is located on a small tubercle on the ventral floor of the vagina.



Lubricating the tip of the catheter with sterile aqueous lubricant.

(Taylor, 2022)

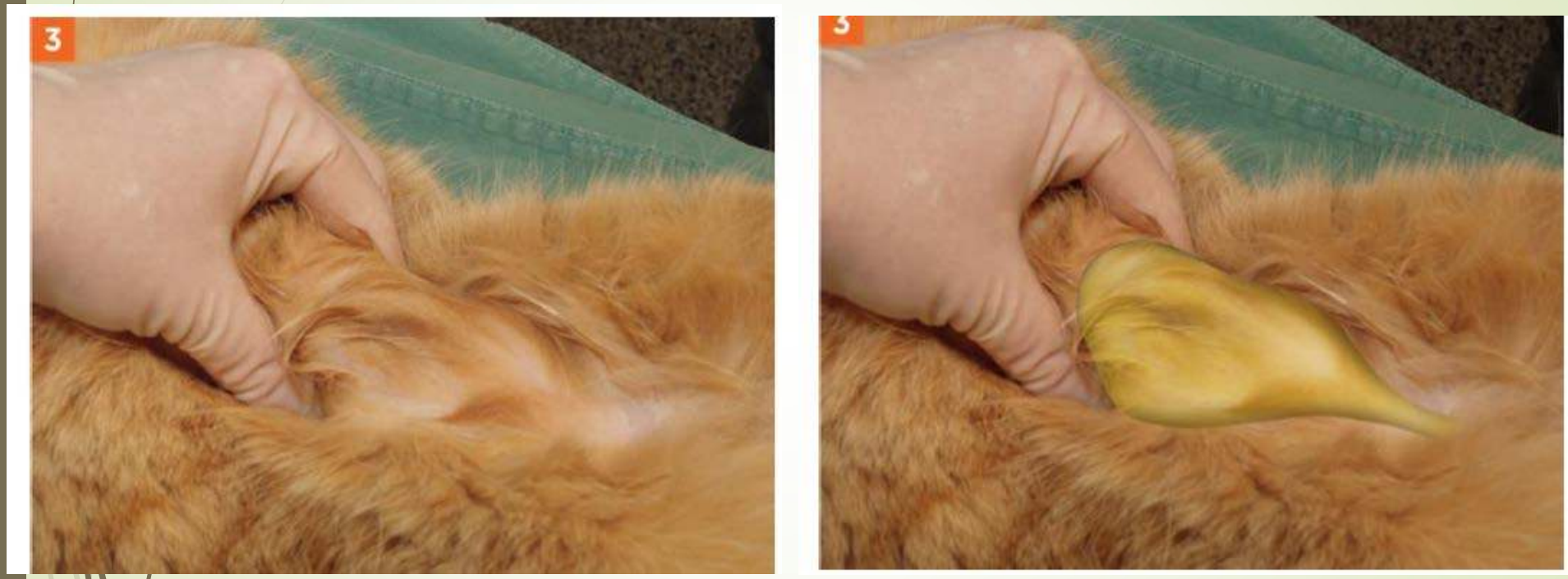
Urinary catheterization for female dog 5



The catheter is inserted into the external urethral orifice and advanced into the lumen of the bladder.

(Taylor, 2022)

Cystocentesis 1



Locating and immobilizing the urinary bladder by gentle palpation.

(Taylor, 2022)

Cystocentesis 2



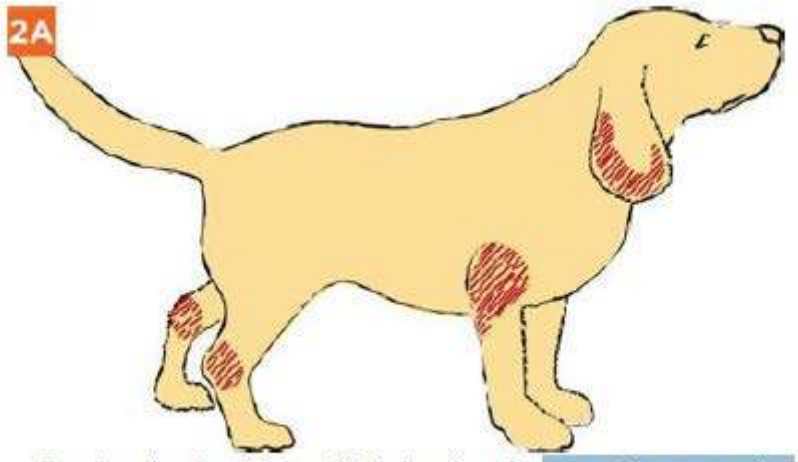
Directing the needle dorsocaudally during cystocentesis.

(Taylor, 2022)

collecting integumentary specimens

Sarcoptic mange

(Taylor, 2022)



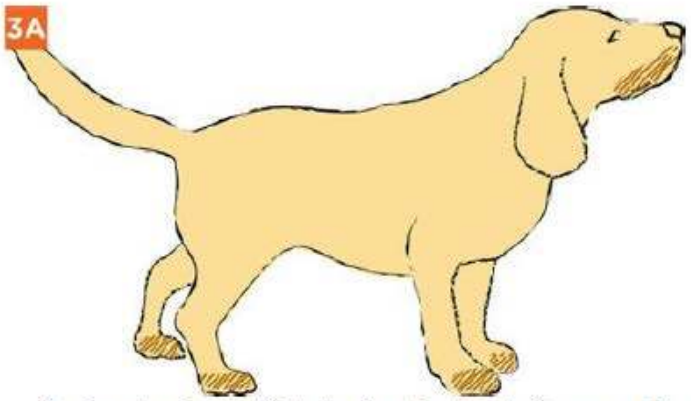
Drawing showing the most likely locations for sarcoptic mange mites.



Alopecia, erythema, and excoriations in a dog with sarcoptic mange.
Source: (Courtesy Dr. Catherine Outerbridge, University of California at Davis.)

Demodex

(Taylor, 2022)



Drawing showing most likely locations for demodectic mange mites.



Facial erythema, scaling, and crusting in a young bulldog with *Demodex* . *Source:* (Courtesy Dr. Catherine Outerbridge, University of California at Davis.)



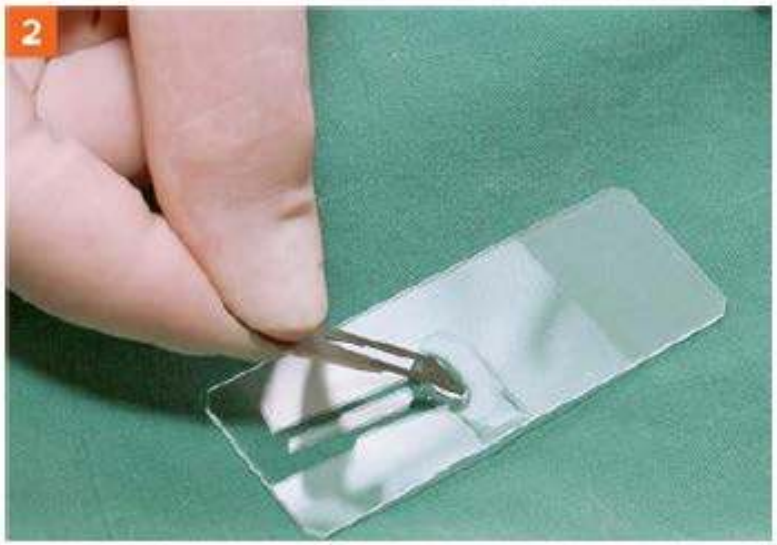
Periocular alopecia, erythema, and scaling in a golden retriever with *Demodex* . *Source:* (Courtesy Dr. Catherine Outerbridge, University of California at Davis.)



Demodex pododermatitis in a dog. *Source:* (Courtesy Dr. Catherine Outerbridge, University of California at Davis.)

Skin scraping

(Taylor, 2022)



Dipping the dull end of a scalpel blade in mineral oil.





The skin is scraped until serum oozes and drops of capillary blood appear.

Skin scraping

(Taylor, 2022)



The scraped material is placed on a microscope slide in a drop of mineral oil and a coverslip applied.

Skin scraping



Microscopic image of sarcoptic (*Sarcoptes scabiei*) mange mite. *Source:* (Courtesy Dr. Klaas Post, University of Saskatchewan.)



Microscopic image of demodectic (*Demodex canis*) mange mites. *Source:* (Courtesy Dr. Klaas Post, University of Saskatchewan.)

(Taylor, 2022)

Cellophane tape method



Equipment required for a cellophane tape preparation.

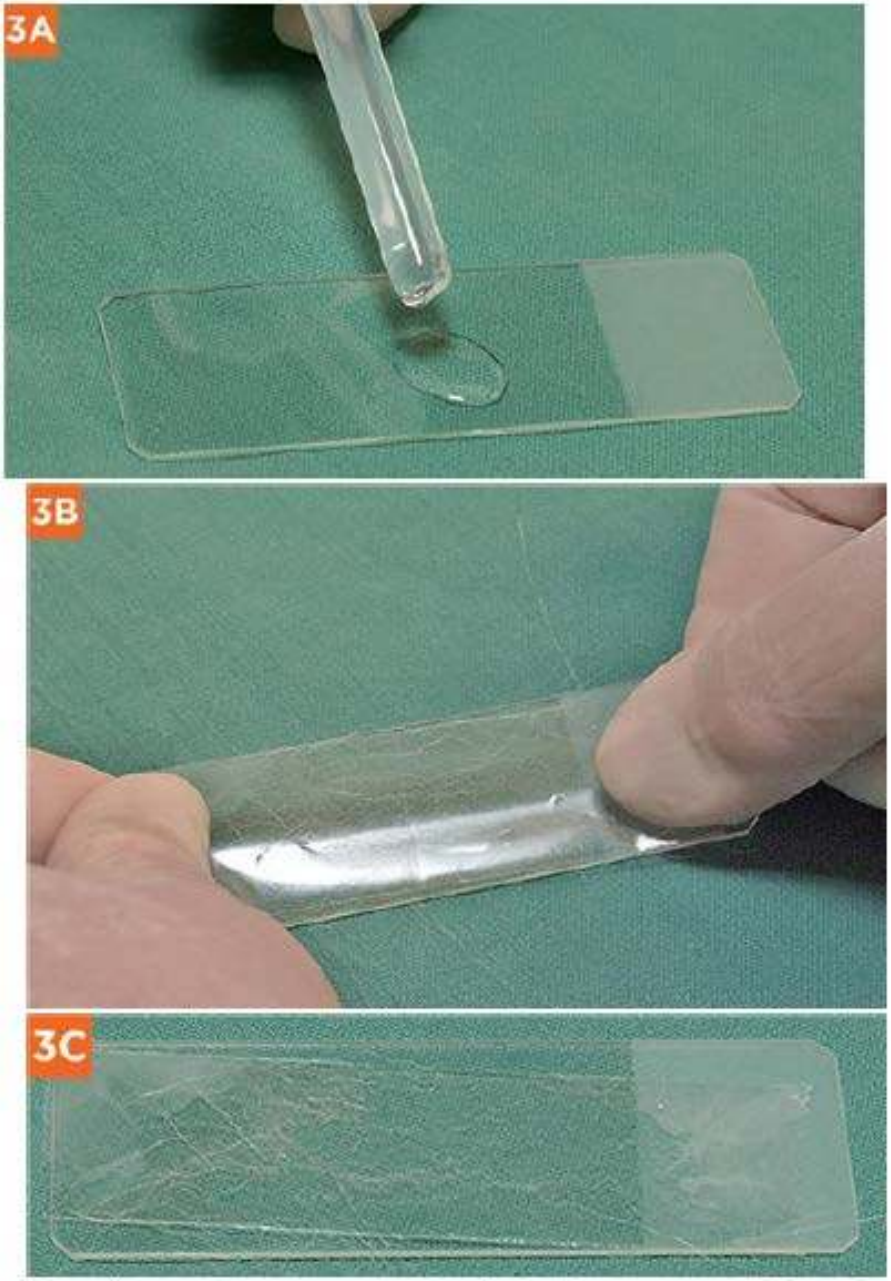


The sticky side of the tape is touched to the hair and skin to collect flakes and debris.

(Taylor, 2022)

Cellophane tape method

(Taylor, 2022)



The tape is applied, sticky-side down, to a microscope slide on top of a drop of mineral oil.

Cellophane tape method



Flaking and pruritus in a young dog with *Cheyletiella* mites ("walking dandruff"). *Source:* (Courtesy Dr. Klaas Post, University of Saskatchewan.)



Cheyletiella yasguri mite identified microscopically using a cellophane tape preparation. *Source:* (Courtesy Dr. Klaas Post, University of Saskatchewan.)

(Taylor, 2022)

Vacuuming



Applying suction to all areas of coat.

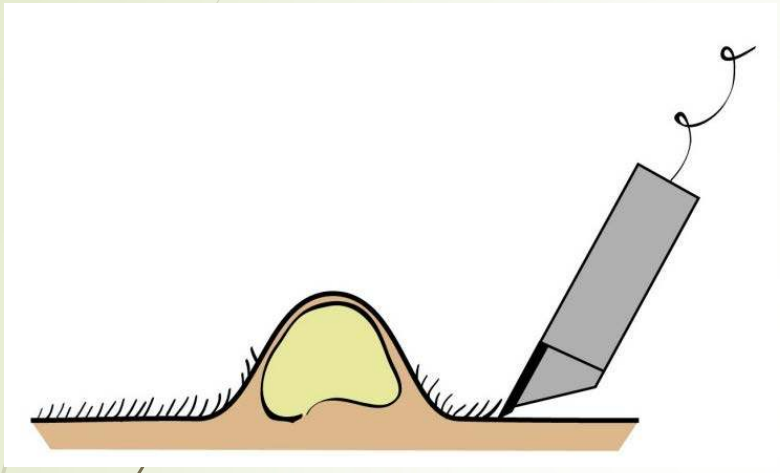


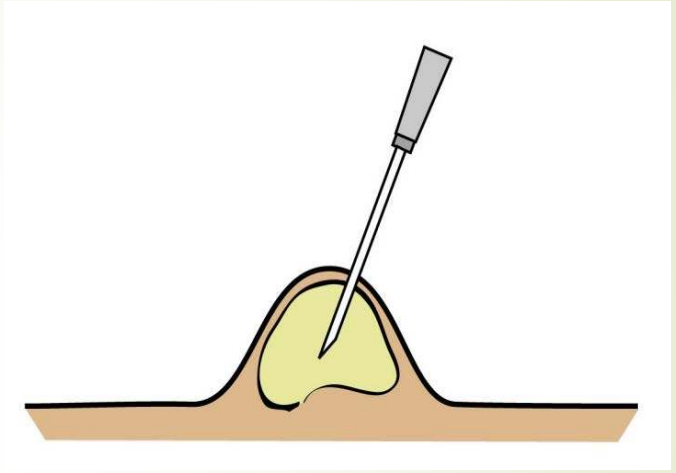
Debris collected by vacuuming is mixed with a few drops of water on a white card. Blood diffuses out of the debris, confirming that the black specks are flea dirt.

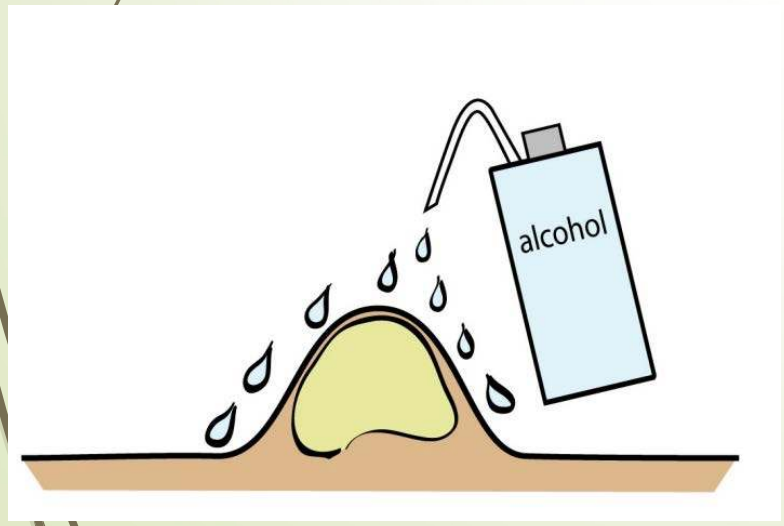
(Taylor, 2022)

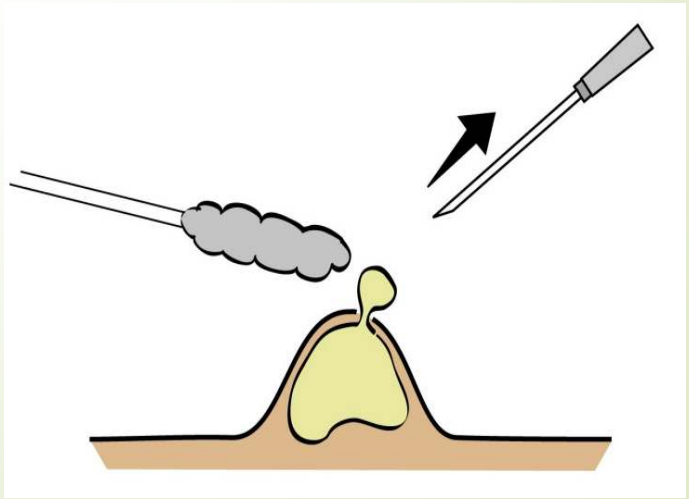
Bacterial culture of skin pustule

(Taylor, 2022)









Ascitic fluid and pleural effusion

Abdominocentesis 1

- 腹腔出現積液
- 針刺位置：腹正中線臍後方2-3cm
- 若只有少量積液，檢查結果多為陰性
  - 可併行診斷性腹膜灌洗：

    灌入20 ml/kg溫生理食鹽水，按摩使液體分布均勻行腹腔穿刺，取至少1ml的盥洗液體
- 細胞學、生化學、微生物學分析

Abdominocentesis 2



A cat restrained in lateral recumbency for abdominocentesis.



(Taylor, 2022)

Abdominocentesis 3

(Taylor, 2022)

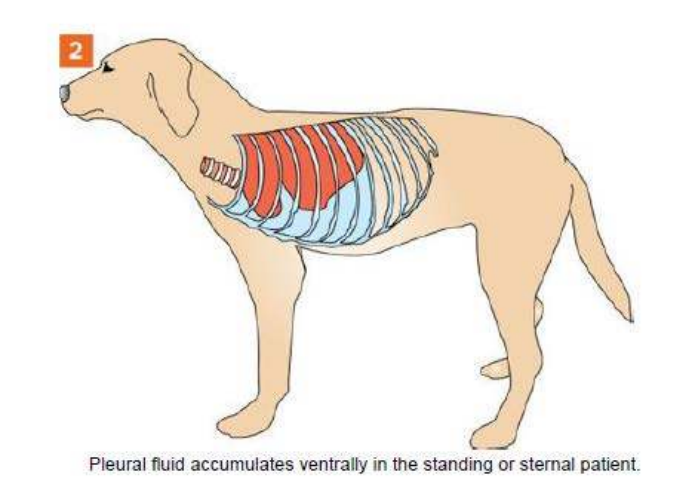


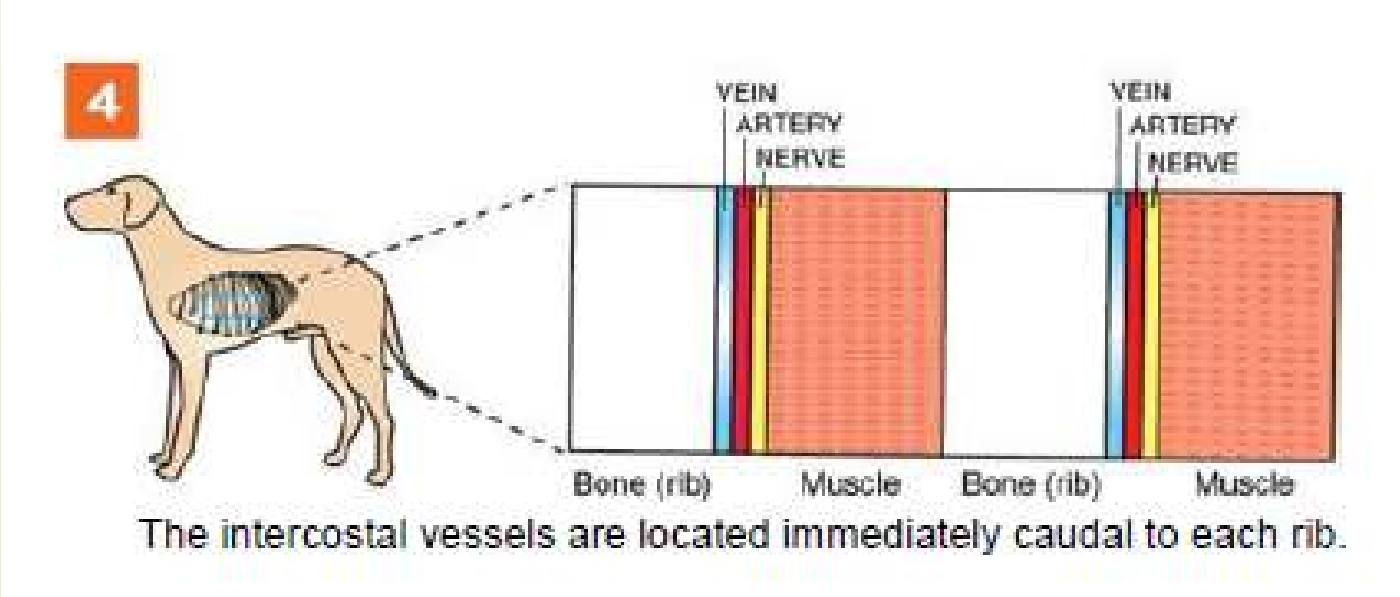
Thoracentesis 1

- 有胸水的犬貓
  - 液體蓄積在胸膜腔
  - 胸膜腔：臟層胸膜與壁層胸膜之間
- 呼吸快而淺，腹側心音與肺音模糊
- X光檢查
- 穿刺位置
  - 動物站立或正趴，使液體聚積腹側
  - 配合理學與X光檢查找出胸水含量與位置
  - 在6^{th}- 7th- 8^{th}肋間，肋軟骨交接處正上方，進針
- 細胞學、生化學、微生物學分析

Thoracentesis 2

(Taylor, 2022)





Thoracentesis 3



Dog restrained while standing for thoracentesis.



Collection of a modified transudate from the pleural space of a dog with right-heart failure.

(Taylor, 2022)

Major references

- Taylor, S. M. 2022. Small animal clinical techniques. Elsevier.
- 賴昱璋翻譯，劉品辰審閱。2022。小動物臨床技術，第三版。Taiwan Elsevier.

臨床暨分子醫學研究室
誠徵研究生

- 臨床醫學與分子診斷學的結合性研究
- 使用創新的分子診斷技術快速診斷犬神經行為病原性的疾病、犬與貓的呼吸道病原性的疾病
- 結合病史與臨床症狀，作流行病學的統計分析
- 參住院醫師與的實習
- **提供研究生獎助學金** (於非住院醫師期間)
- 歡迎欲參加研究生甄試的同學與老師聯絡：
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